

CHSH-Form Values Above 2 for Standard Fibonacci Anyons, and the Non-Topological Origin of Finite-Size Sector–CHSH Mutual Information in $\mathbb{Z}_2$ Lattice Gauge Theory

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We quantify the mutual information $I(T; S)$ between topological sector labels T and CHSH values S in two-dimensional lattice models. Throughout, $I(T; S)$ denotes a classical Shannon mutual information between a global discrete label and a scalar measured value; it is not an entanglement entropy between spatial regions. In the $\mathbb{Z}_2$ lattice gauge theory on a 16×16 torus at inverse temperature $\beta = 2.0$ we measure $I(T; S) = 0.0173$ bits after subtracting the null level, 95% interval $[0.0159, 0.0187]$, significant above 3σ in all 21 seeds against a circular-shift null whose false-positive level we measure at 0.05 of 21. Decomposing the label shows the association is carried entirely by its excitation-number component ($+0.0174$ bits on the gauge-invariant route, 21 of 21 above 3σ); the Wilson-loop component never dominates at any coupling examined, and its own contribution is bounded at 1.2×10^{-4} bits at $\beta = 1.0$, where the label carries 1.9995 of 2 bits of entropy and 3940 of 4000 measurements are effectively independent. The origin of this mutual information is therefore not topological. A finite-size scaling analysis indicates that it decays toward zero with system size. In the quantum toric code via exact diagonalization, abelian $\mathbb{Z}_2$ anyons produce no local CHSH correlations at system sizes $L \geq 3$, consistent with the inaccessibility of abelian topological entanglement to local Pauli operators. We systematically test three classical mechanisms for CHSH values above 2; none exceeds $|S| = 2$ in the implementations studied. Most significantly, within an idealized TQFT model we find that the two-generator d_{1b} circuit model of Fibonacci anyons produces CHSH-form values above 2. With the standard Fibonacci F and R data ($\delta = 0$), the maximum over all 4096 sequences of length $L = 12$ is $|S| = 2.7334$ (96.64% of the Tsirelson bound), for the sequence ABBAABABBAAB. Under a phase deformation δ of these data, the maximum on a 50-point δ -grid is $|S| = 2.8114$ (99.40%) at $\delta \approx 1.44\pi$, for the sequence ABABABABABAB; as a grid maximum it is a lower bound on the maximum over δ . Generic $\delta \neq 0$ breaks the Yang–Baxter/hexagon consistency of the braid representation, so that value characterizes a deformed representation rather than the standard category. The CHSH-form values depend strongly on δ : with fixed measurement settings (from a numerical search at $\delta \approx 1.44\pi$), $|S|$ for that sequence ranges from near zero to 2.72 (above 2 at only 14% of the 200 δ values sampled), while settings adapted to each δ give $|S| > 2$ at 99.5% of them.

Keywords: CHSH inequality, topological order, mutual information, lattice gauge theory, toric code, Fibonacci anyons, Ising anyons, Bell correlations, non-Abelian braiding

I. INTRODUCTION

The violation of Bell inequalities [1] confirmed experimentally by Aspect et al. [2] and in loophole-free tests [3, 4] establishes that no local hidden-variable theory reproduces all quantum predictions. The standard conclusion is fundamental nonlocality, though alternatives exist.

A natural question for a topologically ordered system is whether its internal structure carries information about Bell-type measurement outcomes. The natural candidate is the topological sector. We ask: *does the topological sector of a 2D lattice system carry measurable information about CHSH correlation measurements?* We quantify this using mutual information $I(T; S)$ tested against permutation and circular-shift surrogates, and assess robustness across independent random seeds. The empiri-

cal answer is negative for the $\mathbb{Z}_2$ sector in the thermodynamic limit (Sec. III B).

Our investigation is motivated by the question of whether a 2D manifold with nontrivial topology could provide structured correlation between topological and measurement degrees of freedom [5]; we do not assume that it does, and treat it as an empirical question. While our classical lattice simulations ($\mathbb{Z}_2$, XY) do not achieve Bell violation, the d_{1b} circuit model built from the Fibonacci F - and R -matrices does exhibit CHSH-form values above 2 (Sec. III G), and the classical models establish that topological sectors and CHSH correlations are *not statistically independent* in the $\mathbb{Z}_2$ gauge theory — a necessary condition for any topological mechanism.

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II. MODELS AND METHODS

A. $\mathbb{Z}_2$ Lattice Gauge Theory

Our primary model is the $\mathbb{Z}_2$ gauge theory [8] with Ising variables $\sigma_e \in \{+1, -1\}$ on edges of an $L \times L$ square lattice with periodic boundary conditions:

$$H = -K \sum_p B_p, \quad B_p = \prod_{e \in p} \sigma_e. \quad (1)$$

Topological sectors are labeled by noncontractible Wilson loops $(W_x, W_y) \in \{+1, -1\}^2$. Anyonic excitations correspond to plaquettes with $B_p = -1$. We employ two measurement strategies: (i) gauge-fixed horizontal Wilson lines, where vertical directions are excluded as gauge artifacts under tree gauge-fixing, and (ii) gauge-invariant plaquette-strip observables — products of adjacent plaquette operators along horizontal, vertical, and rectangular paths — which require no gauge fixing and serve as an independent consistency check.

B. XY Model

As an exploratory model, we simulate the classical XY model with continuous angles $\theta_i \in [0, 2\pi)$ on the same torus geometry. Topological sectors are integer winding numbers (n_x, n_y) . Vortex-antivortex defects are identified by plaquette winding numbers. We use Metropolis Monte Carlo with single-site updates.

C. Quantum Toric Code

We perform exact diagonalization of the Kitaev toric code [9] on $L \times L$ tori ($L = 2, 3$):

$$H = - \sum_v A_v - \sum_p B_p \quad (2)$$

with vertex operators $A_v = \prod \sigma^x$ and plaquette operators $B_p = \prod \sigma^z$. Ground states in each of four topological sectors are constructed via projector method. Anyon pairs are created by string operators. CHSH correlations use $M(\theta) = \cos \theta \sigma^z + \sin \theta \sigma^x$.

D. CHSH Measurement Protocol

For classical models, detectors are placed at $A = (L/4, L/4)$ and $B = (3L/4, 3L/4)$ with separation $L/2$ on the 16×16 torus. Measurement outcomes: $A(a) = \text{sign}(\cos(a - \theta_A))$, $B(b) = \text{sign}(\cos(b - \theta_B))$, where θ_A and θ_B denote the orientation of the local field configuration sampled at the respective detector site (for the $\mathbb{Z}_2$ model the four Wilson-line directions realize the settings, with the ± 1 line outcome corresponding to $\theta \in \{0, \pi\}$; for the

XY model, θ is the local spin angle). CHSH settings: $a_1 = 0$, $a_2 = \pi/4$, $b_1 = \pi/8$, $b_2 = -\pi/8$. The CHSH value is $S = E(a_1, b_1) - E(a_1, b_2) + E(a_2, b_1) + E(a_2, b_2)$.

E. Mutual Information and Significance Testing

We compute $I(T; S) = \sum p(t, s) \log_2 \left[\frac{p(t, s)}{p(t)p(s)} \right]$ with S discretized into 20 equal-width bins. Significance is assessed via permutation test (1000–2000 shuffles). The z -score $\sigma = (I_{\text{real}} - \bar{I}_{\text{shuffled}}) / \text{std}(I_{\text{shuffled}})$ quantifies deviation from the null hypothesis of independence. Robustness is assessed across 10 independent seeds. For the corrected engine (Sec. III), significance is assessed against a circular-shift null instead, which displaces one record against the other over all $n-1$ nonzero offsets; it preserves the autocorrelation of both records that a permutation destroys and has no free parameter. Robustness is then assessed across 21 seeds.

III. RESULTS

A. $\mathbb{Z}_2$ Gauge Theory: Primary Result

Engine. Unless marked as original analysis, all $\mathbb{Z}_2$ values in this section, in the abstract, and in Tables II and III are from the corrected engine described in the disclosure of Sec. IIIB, with significances against the circular-shift null. The original analysis on the superseded engines comprises Table I, Tables IV and V, and the $\mathbb{Z}_2$ rows of Table VII; their captions say so.

Robustness across seeds. Table I shows $I(T; S)$ across 11 independent Monte Carlo runs (10 systematic + original seed 42) at $\beta = 2.0$ on a 16×16 torus with 4000 measurements each. The original discovery run (seed 42) yielded $I = 0.011$ bits at 6.6σ . These per-seed values are the original analysis on a superseded engine with a permutation null. The corrected engine reproduces the raw mean over 21 seeds (0.0150 bits); after subtracting the circular-shift null the association is 0.0058 bits (95% interval $[0.0038, 0.0078]$). The per-seed significance does not carry over: 10 of 21 seeds exceed 2σ and 4 of 21 exceed 3σ , against 9 of 10 above 2σ in the table.

Decomposition of the sector label. Table II decomposes the sector label into Wilson-loop and anyonic components across temperatures (Fig. 1). At $\beta \leq 1.5$ no component of the label is associated with S : every 95% interval includes zero. The isolated seeds above 2σ — the Wilson-loop component in 1 of 21 seeds at $\beta = 1.0$, the excitation-number component in 2 of 21 at $\beta = 1.5$ — are compatible with the false-positive rate we measure for the null itself in those cells (0.6 and 0.8 expected of 21). At $\beta = 2.0$ and 2.5 the excitation-number component carries the association (6 and 15 of 21 seeds above 3σ against the circular-shift null; 8 and 17 of 21 above 2σ), and it does so at every coupling

examined — the Wilson-loop component never dominates. Its own contribution is bounded rather than absent: at $\beta = 1.0$, where the sector label carries 1.9995 of 2 bits of entropy and 3940 of 4000 measurements are effectively independent, it is -0.000055 bits with a 95% interval of $[-0.00020, +0.00012]$, an upper bound of 1.2×10^{-4} bits. The one interval that excludes zero is $\beta = 2.5$ ($[+0.00092, +0.00963]$) — the cell with the poorest resolution among the couplings with a stated bound, 73 of 4000 independent. The apparent contribution grows as the resolution falls from 3940 to 12 of 4000, which is the signature of a residual estimator bias rather than of physics. At $\beta = 3.0$ the label itself is close to freezing (1.484 of 2 bits of entropy, 12 of 4000 independent) and no bound is claimed there.

The loss of resolution has an established explanation, and it is not specific to our setting. In the literature on the thermal stability of topologically ordered systems, the physical engine behind the loss of correlations is the proliferation of topological defects at any finite temperature [28], and the relaxation time this produces is bounded by a constant that grows exponentially with the ratio of the excitation gap to the temperature and does not increase with system size [29]. That the defects must also move is explicit there: the dynamics is described in terms of diffusing anyons, and penalizing their motion lengthens the memory lifetime [31], with creation and hopping carrying separate energy costs [32]. That a two-dimensional stabilizer code cannot protect the sector at all is a separate statement, independent of these rates [30].

The integrated autocorrelation time we measure for the Wilson-loop pair (W_x, W_y) is structurally the same quantity as the autocorrelation time of the nonlocal topo-

logical observables studied there. The dynamics is not the same: those results concern a quantum memory coupled to a thermal bath, ours a classical Metropolis chain on gauge configurations. We therefore quote the established picture and treat our setting as a classical Monte Carlo analogue of it rather than an instance of it. What this work adds is the measurement in this setup — the resolution of the sector label at each coupling, and the anticorrelation between the apparent signal and that resolution.

A related observation has been reported independently in a $U(1)$ Schwinger-model setting: there, mutual information in the flux sector can remain large while no local fragment can convert it into classical information, so that topological flux records are globally distinguishable but invisible to local matter fragments [33]. Our bound is what that statement predicts, in a different system and with a different pair of variables: their mutual information is between a system and environment fragments, ours between a global label and a scalar measured value; their model is $U(1)$ and solved by exact diagonalization, ours is $\mathbb{Z}_2$ and sampled by a Metropolis chain; and their direction is large but locally unreadable, ours bounded near zero.

Important caveat for $\beta \geq 2.5$: In this regime excitations are rare and the sector label changes slowly: over the 21 seeds of Table II the mean anyon number is 1.6 at $\beta = 2.5$ and 0.3 at $\beta = 3.0$. The label is not frozen, and the mutual information does not vanish; it is carried by the excitation-number component (Table II). The superseded engine, on which the original version of that table rested, showed a stronger effect, which we call its *frozen regime*: near-zero anyon numbers and, at $\beta = 3.0$ and 4000 measurements, a CHSH value that was exactly constant, so that the mutual information vanished identically. A characterization deposited with this record

TABLE I. Mutual information $I(T; S)$ in $\mathbb{Z}_2$ gauge theory ($L = 16$, $\beta = 2.0$). Systematic seeds 10–19 plus original discovery run. Original analysis on a superseded engine with a permutation null (Sec. III B); corrected values in the text. The last row gives the mean $\pm$ population standard deviation across the ten systematic seeds.

Seed	I (bits)	σ	p -value
42 (original)	0.0109	6.6	< 0.001
10	0.0397	31.6	< 0.001
11	0.0176	11.5	< 0.001
12	0.0178	14.8	< 0.001
13	0.0099	4.7	< 0.001
14	0.0301	26.2	< 0.001
15	0.0061	2.6	0.005
16	0.0121	8.2	< 0.001
17	0.0035	1.0	0.15
18	0.0075	4.6	< 0.001
19	0.0073	3.1	< 0.001
Mean (10 syst.)	0.015 ± 0.011	10.8 ± 10.0	9/10 $> 2\sigma$

TABLE II. Decomposition of the sector label at $L = 16$ for the gauge-fixed Wilson-line measurement (the gauge-invariant plaquette-strip decomposition at $\beta = 2.0$ is given in the text): corrected engine, 21 seeds {42, 10–29}, 4000 measurements per run. Entries: seed mean of $I(T; S)$ after subtracting the circular-shift null (bits), with the number of seeds above 3σ against that null in parentheses; Anyons: mean anyon number. The mutual information of the full label is not the sum of those of its components. Recomputed from the deposited per-seed values (Data and Code Availability). The original analysis of this table (superseded engine, five seeds, permutation null) showed exactly vanishing entries at $\beta = 3.0$; that was a property of the superseded engine (Sec. III B).

β	Anyons	I_{full}	I_{Wilson}	I_{anyons}
1.0	30.5	-0.0001 (0/21)	-0.0001 (0/21)	0.0000 (0/21)
1.5	12.2	0.0005 (0/21)	0.0000 (0/21)	0.0000 (1/21)
2.0	4.7	0.0058 (4/21)	0.0005 (2/21)	0.0028 (6/21)
2.5	1.6	0.0498 (15/21)	0.0049 (3/21)	0.0354 (15/21)
3.0	0.3	0.0366 (18/21)	0.0038 (4/21)	0.0416 (21/21)

(`data/frozen_regime_characterization.py`) quantifies this frozen regime for both superseded engines on the 40 000-measurement runs and, for the legacy engine at $\beta = 3.0$, additionally at the 4000-measurement scale of the original analysis. On the long runs, sector transitions of the legacy engine are rare at $\beta = 3.0$ (mean block length ≈ 315 measurements) but frequent at $\beta = 2.5$ (roughly 900–10 500 per 40 000 measurements; the second engine shows 214–237 at $\beta = 3.0$ on the same scale), so in those runs the elevated $\beta = 2.5$ mutual information is not a rarity effect. Where transitions are rare, the sector label stays constant over long blocks of the sampled series, and the permutation null — which destroys exactly this temporal block structure — becomes very narrow, so that large significances arise at small $I(T; S)$. Measured directly by pairing records of independent seeds, the permutation null produces 5.0 of 5 false positives at $\beta = 2.5$ and 1.0 ± 0.2 of 21 on the gauge-invariant route, against 0.2 of 5 and 0.05 of 21 for the circular-shift null. All significances reported here use the latter. Rarity itself is quantified by the integrated autocorrelation time of the full sector label (`tau_int_labels` in the deposited record), which rises from 0.5 at $\beta = 1.0$ to 229 at $\beta = 3.0$. Three further autocorrelation times are recorded alongside it, for the Wilson-loop pair and for the excitation-number component; at $\beta = 3.0$ the four span a factor of four, from 61 to 244, so a value of this kind carries no meaning without the definition it was computed under. The conditional-variance ratio $\text{Var}(S|T)/\text{Var}(S) \approx 0.93$ – 1.0 on the long runs of the superseded engines, where $\text{Var}(S) > 0$, shows that S is *not* a near-deterministic function of the sector label under long sampling. The historical single-run value $I(T; S) = 0.320$ recorded in Table VII predates this protocol: it is not reproduced by the deposited seed set, and the $|S|$ value recorded with it differs from the constant value of the superseded engine’s runs, so it is kept as history only. The elevated value at $\beta = 2.5$ is not evidence for a topological mechanism: the decomposition attributes it to the excitation-number component of the label, and the Wilson-loop component alone contributes $+0.005$ bits with 3 of 21 seeds above 3σ . The earlier reading of this regime as a near-frozen anyon sector rested on the superseded engine, which produced 0.04 excitations here instead of 1.6. At $\beta \sim 2.0$ the association is carried entirely by the excitation-number component of the label. Measured separately on the same runs over 21 seeds on the gauge-invariant plaquette-strip route, that component alone yields $+0.0174$ bits (21 of 21 seeds above 3σ), while the Wilson-loop component alone yields $+0.000033$ bits with a 95% interval of $[-0.00028, +0.00043]$ — consistent with zero. The holonomies of the sector are not the carrier.

Gauge-invariant measurement. To rule out gauge-fixing artifacts, we repeat the $\beta = 2.0$ analysis using gauge-invariant observables: products of adjacent plaquette operators along horizontal, vertical, and rectangular paths (length 4). These require no gauge fixing and are manifestly physical. Table III compares the two

measurement strategies.

On gauge-invariant observables the association is present in every seed: $I(T; S) = 0.0173$ bits after subtracting the circular-shift null, significant above 3σ in all 21 seeds at a measured false-positive level of 0.05 of 21. On the gauge-fixed Wilson lines it is present in the seed mean (0.0058 bits) but exceeds 3σ in only 4 of 21 seeds. The two rows measure different observables, and their sizes are not compared. This confirms that the measured association is not an artifact of the gauge-fixing prescription (its finite-size scaling, and whether it admits a non-finite-size physical reading, are examined in Sec. III B).

B. Finite-Size Scaling of $I(T; S)$

Motivation. The $I(T; S) = 0.015 \pm 0.011$ bits (standard deviation across seeds) reported as the original analysis in the previous subsection is a measurement at fixed lattice size $L = 16$ and fixed temperature $\beta = 2.0$. A natural question, raised by gauge-theory generalities, is how this signal scales with L : does it persist toward the thermodynamic limit, or is it a finite-size effect of the $L = 16$ system used for the original measurement? We report here a dedicated scaling study at fixed $\beta = 2.0$. The aim is not to retract the $L = 16$ value but to embed it in a scaling sequence.

Methods. We measure $I(T; S)$ at $L \in \{8, 16, 32, 48, 64\}$, $\beta = 2.0$, with 10 independent random-seed runs per (L, engine) cell. Two independent Monte Carlo engines are used, sharing the single-spin-flip Metropolis update rule but differing in sweep order: (i) the legacy engine `z2.sim.v05.py` (same as for the original analysis of Table I) sweeps the link variables in row-major order; (ii) a newly written vectorized engine `z2.vec.py` updates the link variables in a checkerboard order. The two engines are com-

TABLE III. Comparison of measurement strategies at $\beta = 2.0$ ($L = 16$), corrected engine, seeds {42, 10–29}. I : seed mean after subtracting the circular-shift null, with 95% bootstrap interval; σ : mean and standard deviation across seeds of the per-seed significance against that null. Original analysis in the footnote.

Method	I (bits)	σ	Seeds $> 3\sigma$
Gauge-fixed	0.0058	2.2 ± 2.0	4/21
Wilson lines	[0.0038, 0.0078]		
Gauge-inv.	0.0173	10.5 ± 1.7	21/21
plaq. strips	[0.0159, 0.0187]		

^a Original analysis (superseded engine, permutation null; standard deviations across seeds are population values except the gauge-invariant significance spread, 11.4, which is a sample value, as in the original analysis): gauge-fixed 0.015 ± 0.011 bits, mean permutation significance 10.8 ± 10.0 , 8 of 10 seeds above 3σ ; gauge-invariant 0.018 ± 0.011 bits, mean permutation significance 14.2 ± 11.4 , 21 of 21.

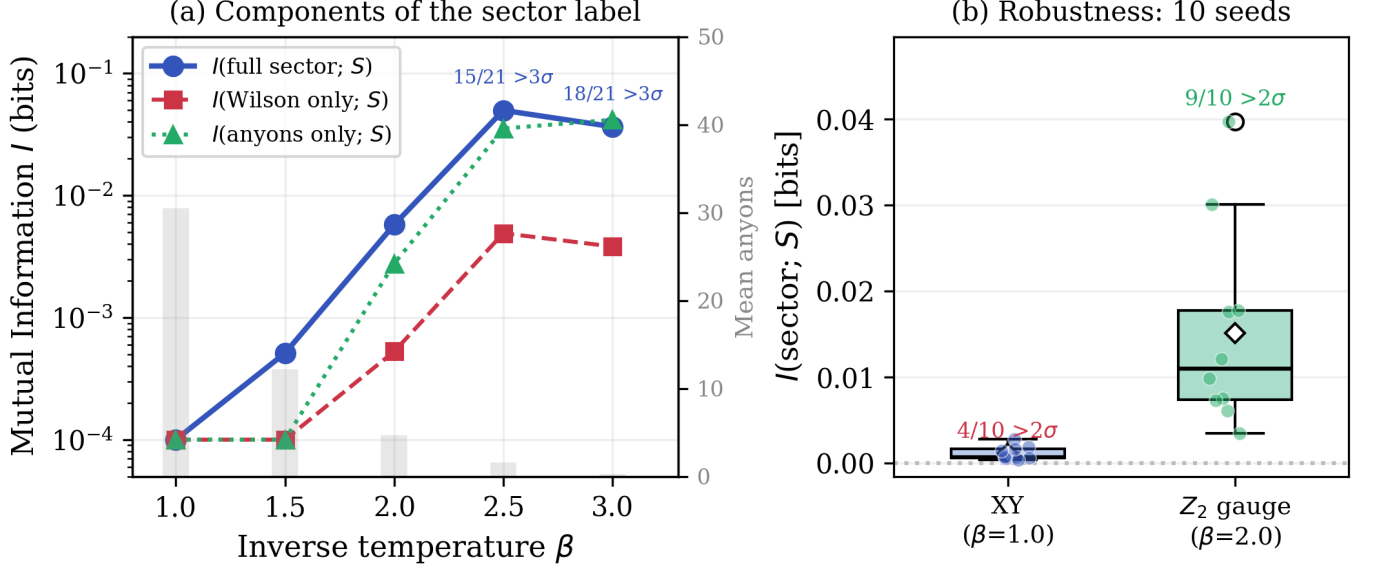


FIG. 1. (a) Mutual information vs. inverse temperature for the components of the sector label (corrected engine, 21 seeds, circular-shift null subtracted; Table II). Gray bars: mean anyon count. (b) Seed variation: 10 independent runs per model (original analysis; for Z_2 the seeds of Table I). Values in (a) are clipped at 10^{-4} for the logarithmic axis.

pared against the analytic single-plaquette expectation $\langle B_p \rangle = \tanh(\beta)$ (Wegner duality limit, exact for the 2D Z_2 gauge model at the level of single-plaquette averages). Statistical significance per cell is reported as $\sigma = (\bar{I}_{\text{real}} - \bar{I}_{\text{shuffled}}) / \sqrt{\text{SEM}_{\text{real}}^2 + \text{SEM}_{\text{shuffled}}^2}$, the difference of the seed means divided by their combined standard error across the 10 seeds, with 2000 permutation shuffles per seed. Three model classes are pre-registered for the L -dependence fit: a pure power law $M_2 : aL^b$; a power law with nonzero asymptote $M_{2c} : aL^b + c$; and a constant-asymptote exponential $M_{\text{exp}} : ae^{-bL} + c$. Model selection is by Bayesian information criterion (BIC), with $\Delta\text{BIC} > 6$ used as the threshold for decisive preference among the three.

Results — per-cell measurements. Table IV reports $I_{\text{mean}} \pm \text{SEM}$ over the 10 seeds per cell, together with the standardized significance against the permutation null. The legacy-engine value at $L = 16$, $I = 0.0149 \pm 0.0036$ bits, reproduces the original-analysis mean of Table I (0.015 ± 0.011 bits, standard deviation across seeds) to within 0.03σ .

BIC model comparison. Three model classes are pre-registered for the L -dependence of the sector-shuffle-corrected $I(T; S)(L)$ at $\beta = 2.0$ (the raw $I(T; S)$ minus the mean $I(T; S)$ of sector-shuffled surrogates, i.e., the permutation-null-subtracted signal reported per cell in `data/stufe2.analyse.json`): a pure power law $M_2 : aL^b$ (2 parameters); a power law with nonzero asymptote $M_{2c} : aL^b + c$ (3 parameters); and a constant-asymptote exponential $M_{\text{exp}} : ae^{-bL} + c$ (3 parameters). Table V reports BIC and χ^2 for each model and each engine.

Engine consensus. M_{exp} minimizes BIC in both engines (Fig. 2). The strength of the preference differs:

in the vectorized engine $\Delta\text{BIC}_{\text{best vs 2nd}} = 7.18$, above the pre-registered decisive threshold; in the legacy engine the gap is 3.15, positive but sub-decisive. Given the sub-decisive gap in the legacy engine, the engine-consensus carries the model-selection conclusion, not either engine alone — M_{exp} is the *preferred* model among the pre-registered set, not a uniquely determined one. The fitted exponential rate is consistent across engines: $b = 0.125 \pm 0.020$ (v05) and $b = 0.140 \pm 0.024$ (vec), corresponding to a characteristic length scale of $1/b \approx 7$ –8 lattice spacings. The fitted asymptote c is statistically indistinguishable from zero in both engines: $c =$

TABLE IV. Mutual information $I(T; S)$ vs. lattice size L at fixed $\beta = 2.0$, 10 seeds per cell: original analysis on the two superseded engines (legacy v05, vectorized vec; disclosure in the text). σ is the difference of the seed means of $I(T; S)$ and of its permutation surrogates (2000 shuffles per seed), divided by their combined standard error across seeds. The corrected engine’s values are given in the text.

L	v05: I (bits)	σ	vec: I (bits)	σ
8	0.0372 ± 0.0108	3.5	0.0191 ± 0.0029	6.6
16	0.0149 ± 0.0036	4.1	0.00694 ± 0.00144	5.1
32	0.00235 ± 0.00069	2.5	0.000544 ± 0.000628	0.6
48	0.000091 ± 0.000315	0.3	$-3.5 \times 10^{-6} \pm 6.1 \times 10^{-5}$	0.2
64	0.0000857 ± 0.0000461	0.6	$2.9 \times 10^{-7} \pm 2.1 \times 10^{-5}$	0.5

^a All cell-level values reproducible from `data/stufe2.analyse.json` (included in this release), which `data/stufe2.analyse.py` regenerates from the raw run `data/stufe2.results.json`; seeds 10–19 systematic. The σ column is the field `sig_minus_scrambled` of that file.

$(0.62 \pm 8.8) \times 10^{-5}$ (v05) and $c = (0.10 \pm 3.2) \times 10^{-5}$ (vec), each with $|c|/\text{SEM}(c) < 0.1$. Two qualitative conclusions follow that are robust across both superseded engines: (i) $I(T; S)(L)$ decays faster than any power law in the tested range, and (ii) the asymptote is data-supported (not extrapolated) to be indistinguishable from zero. The corrected engine (10 seeds per cell, circular-shift null) gives 0.0248 [0.0143, 0.0360] bits at $L = 8$ and 0.0051 [0.0026, 0.0074] bits at $L = 16$, and 95% intervals that include zero from $L = 32$ on; conclusion (ii) therefore holds on all three engines, while the model comparison behind conclusion (i) rests on the superseded engines. The absolute $L = 16$ value is, however, *not* engine-robust — see disclosure below.

Superseded engines — disclosure. The two engines do not agree on the absolute $I(T; S)$ value at fixed (L, β) . At $L = 16$, $\beta = 2.0$, the legacy engine reports $I = 0.0149 \pm 0.0036$ bits while the vectorized engine reports $I = 0.00694 \pm 0.00144$ bits, a factor of ~ 2.15 . The two engines also differ in their reproduction of the analytic single-plaquette expectation $\langle B_p \rangle = \tanh(\beta)$ (Wegner-duality limit, exact for the 2D $\mathbb{Z}_2$ gauge model at the level of single-plaquette averages): at $\beta = 2.0$ the legacy engine deviates by +2.2% toward colder configurations (original diagnostic), and the deposited validation shows that the vectorized update, too, misses the exact finite-torus value, by +0.66% at $L = 16$. The mechanism has since been isolated. The legacy engine visits the links in a fixed row-major order and accepts every energy-neutral flip deterministically; together these make its Markov chain reducible, so that time averages depend on the starting configuration. The vectorized engine shares the deterministic acceptance and is biased by it. The corrected engine accepts energy-neutral flips with probability 1/2 and otherwise keeps the vectorized update. Against the exact plaquette average of the finite torus, of which $\tanh(\beta)$ is the unconstrained limit, it passes at all four validation points to within 1.2 standard errors, while the same engine without this change misses by 4.3 to 52.6 standard errors. All $\mathbb{Z}_2$ values in Sec. III

TABLE V. Bayesian information criterion (BIC) for three pre-registered L -scaling models of $I(T; S)(L)$ at $\beta = 2.0$, fitted to Tab. IV. χ^2 is the unweighted sum of squared residuals. $\Delta\text{BIC}_{\text{best vs 2nd}}$ above 6 is the pre-registered threshold for decisive preference; 2–6 counts as positive evidence. Original analysis on the two superseded engines; no model comparison was computed for the corrected engine.

Model	v05: BIC	v05: χ^2	vec: BIC	vec: χ^2
M_2	12.33	9.11	15.77	12.55
M_{2c}	8.26	3.44	12.86	8.03
M_{exp} (best)	5.11	0.28	5.69	0.86
$\Delta\text{BIC}_{\text{best vs 2nd}}$	3.15		7.18	

^a Fit parameters and per- L predictions reproducible from `data/stufe2_analyse.json`.

and in Tables II and III are from the corrected engine. The plaquette diagnostic does not, by itself, *select* among the three scaling models above; model selection rests on the BIC comparison of Tab. V. The previously reported $I = 0.015 \pm 0.011$ bits (standard deviation across seeds) at $L = 16$ (Sec. III) remains a correct measurement of the legacy engine’s output, and the corrected engine reproduces it as a raw value (0.0150 bits over 21 seeds; 0.0152 ± 0.0012 bits, standard error, on the scaling seeds). What does not carry over is the per-seed significance attached to it (Sec. III). The vanishing asymptote of the consensus paragraph holds on all three engines.

Conclusion. The subsection has two robust take-aways and one scope-limiting statement. *Robust:* (i) at fixed $\beta = 2.0$, $I(T; S)(L)$ decreases faster than any power law in the tested range $L \in \{8, 16, 32, 48, 64\}$, with the exponential form M_{exp} preferred under the pre-registered set in both engines; (ii) the asymptote is data-supported (not extrapolated) to be statistically indistinguishable from zero, also for the corrected engine. *Engine-specific:* the absolute $L = 16$ value differs by a factor of ~ 2 between the two engines and should be read as engine-specific (legacy: 0.0149 bits, vectorized: 0.00694 bits; the corrected engine gives 0.0152 bits); the qualitative trend is unaffected by this specificity. *Scope.* The methodological question of how a finite-size $I(T; S) > 0$ at $L = 16$ that decays toward zero at larger L should be interpreted — in particular, whether the observed signal admits a non-finite-size physical reading — is beyond the empirical scope of this paper and is deferred to future theoretical work.

C. XY Model: Exploratory

In the XY model at $\beta = 1.0$, seed variation reveals that $I(|n|; S)$ is not robust: only 4/10 runs exceed 2σ ($2/10 > 3\sigma$). Mean: $I = 0.001 \pm 0.001$ bits at $1.5 \pm 1.9\sigma$. The trend $\uparrow |n| \rightarrow \downarrow S$ has correlation $r = +0.03 \pm 0.75$, consistent with zero. We report this as exploratory. The signal may require larger lattices or longer runs to establish in this noisier model.

Vortex-antivortex correlations show Coulomb-like attraction ($g(r=1) = -0.78$) consistent with BKT physics [10], and the CHSH value anti-correlates with defect count ($r = -0.90$), indicating that topological disorder degrades CHSH correlations.

D. Quantum Toric Code

At $L = 2$ (8 qubits), m -anyon pairs yield $|S| = \sqrt{2}$ with the sign flipping between sectors ($I = 0.350$ bits, 1595σ). At $L = 3$ (18 qubits), all single-qubit and Wilson-loop correlations are exactly zero — verified across all 153 qubit pairs and all operator types (σ^z - σ^z , σ^x - σ^x , mixed).

The $L = 2$ result is a finite-size artifact: on a 2×2 torus, measurement qubits simultaneously belong to the

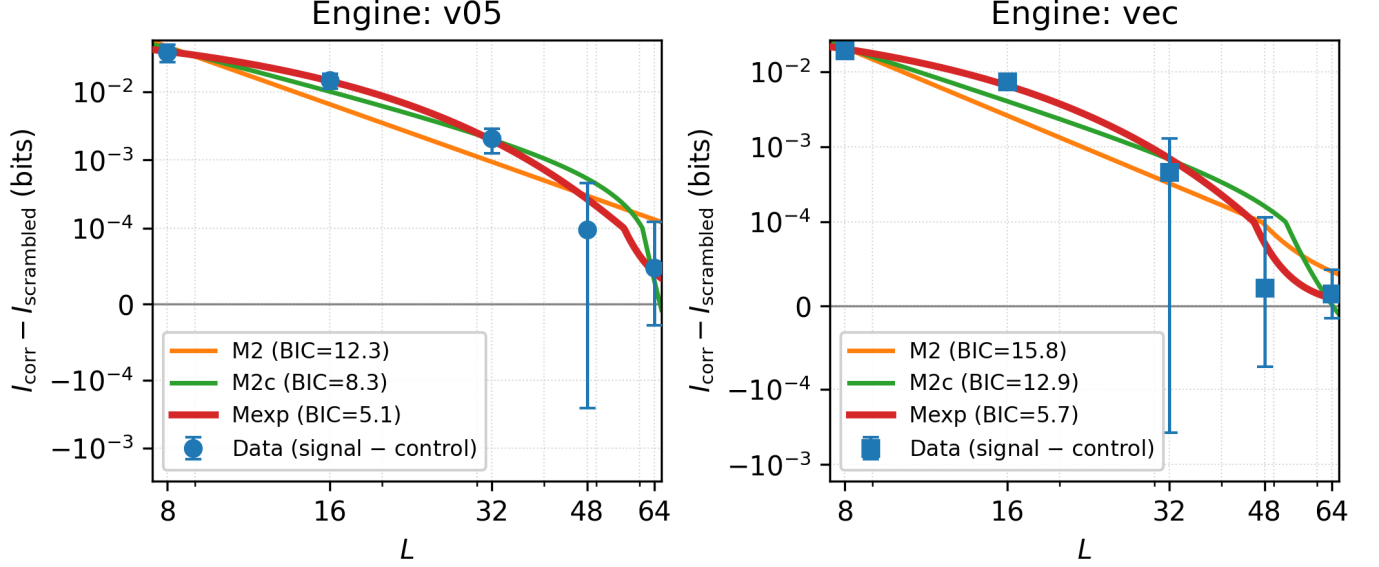


FIG. 2. Finite-size scaling of $I(T; S)$ in 2D $\mathbb{Z}_2$ gauge theory at $\beta = 2.0$. Per-cell measurements (Tab. IV) with two Monte-Carlo engines (legacy v05 = circles, vectorized vec = squares) and the three pre-registered fit curves (Tab. V): pure power M_2 , power-with-asymptote M_{2c} , exponential M_{exp} . M_{exp} is the BIC-preferred model in both engines (decisive in vec, sub-decisive in v05). The engine-consensus conclusion is qualitative: $I(T; S)(L)$ decays faster than any power law in the tested range, with asymptote data-supported to be zero within the measurement precision. The absolute $L = 16$ value is engine-specific (see disclosure in text). Original analysis on the two superseded engines; the corrected engine's per-cell values are given in the text.

string, the plaquettes, and noncontractible loops. At $L \geq 3$, the abelian nature of $\mathbb{Z}_2$ anyons means their entanglement is encoded globally and cannot be accessed by any operator that is a product of Pauli matrices on a contractible subset of edges [11, 12].

E. Systematic Negative Results

We tested three classical mechanisms for Bell violation:

(i) **Measurement-rule modification:** Making $A(a, \lambda)$ depend on the topological sector n via a phase shift $A = \text{sign}(\cos(a - \theta + n\alpha))$ decreases $|S|$ — the coupling adds noise rather than correlation.

(ii) **Statistical-independence violation:** Weighting configurations by $\rho(n|a, b)$ allows S to reach 2.000 exactly but never exceed it. Fine's theorem [13] guarantees $|S| \leq 2$ for deterministic local functions $A(a, \lambda)$ and $B(b, \lambda)$ only when the hidden-variable weighting $\rho(\lambda)$ is *setting-independent*; the present family $\rho(n|a, b)$ is explicitly setting-dependent, so the theorem's hypothesis is not strictly satisfied here. Our result is narrower and empirical: this particular sector-mediated weighting family caps at $S = 2.000$ in the numerics tested, not a general re-derivation of the Fine bound for setting-dependent weightings.

(iii) **Transport phase:** Including a gauge-field phase accumulated along the path between detectors produces the correct angular form (correlation with $-\cos \theta$ at $r = 0.993$) but only 17% of the quantum-mechanical amplitude. Thermal fluctuations wash out the phase coher-

ence.

F. Fourier Analysis

The sector difference $\Delta C(\theta)$ (Fig. 3) shows the dominant Fourier mode is $k = 1$ in all sectors, inconsistent with the prediction that the mode should scale with winding number n . A weak resonance enhancement at $\theta = \pi/n$ for $|n| \geq 2$ (amplitude ratio 1.20–1.38) is observed but not statistically decisive. The even-harmonic structure in raw residuals ($a_2 \gg a_1$) is an artifact of the $\text{sign}(\cos)$ measurement discretization, eliminated by using soft $\langle \cos \rangle$ measurements.

G. Fibonacci Anyons: CHSH-Form Values

Fibonacci anyons have fusion rule $\tau \times \tau = 1 + \tau$ (golden ratio quantum dimension) and no fermion parity superselection. Their braiding generates a dense subset of $\text{SU}(2)$ per logical qubit, making them universal for quantum computation [14]. We simulate two logical qubits encoded in six Fibonacci anyons with the d_{1b} circuit model, a two-qubit circuit whose gates are constructed from the Fibonacci F -matrix

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad (3)$$

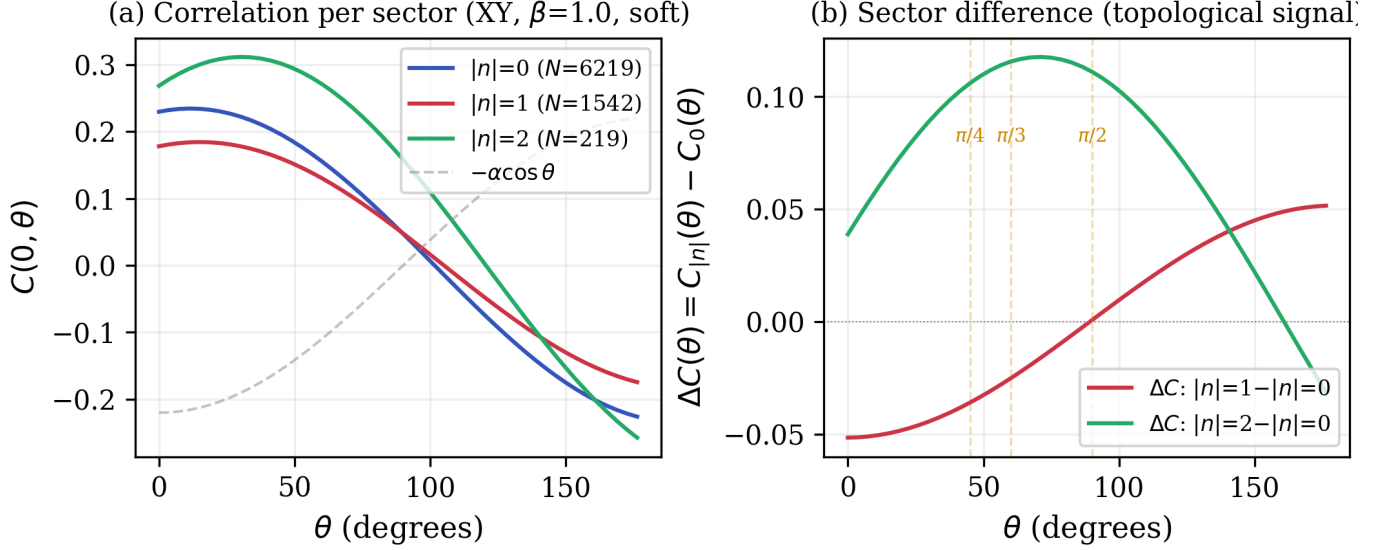


FIG. 3. (a) Correlation $C(0, \theta)$ per winding-number sector in the XY model (soft measurement). (b) Sector difference $\Delta C = C_{|n|}(\theta) - C_0(\theta)$, isolating the topological signal. Dashed lines mark predicted resonance angles π/n .

and R -matrix ($R_1 = e^{-4i\pi/5}$, $R_\tau = e^{3i\pi/5}$). One gate acts on Alice's qubit alone, the other acts on Bob's qubit conditioned on Alice's; the two gates do not satisfy the braid relations. We denote them AM and MB. Braiding alone, with the two commuting braid generators of the same encoding, does not exceed $|S| = 2$ (Ref. [6]).

Table VI shows the maximal CHSH value as a function of sequence length. No enumerated sequence from $L = 3$ to $L = 12$ falls below the classical CHSH bound: each either violates it ($|S| > 2$) or saturates it exactly ($|S| = 2$ to within 10^{-10}). The violating fraction at $\delta = 0$ rises monotonically from $4/8$ at $L = 3$ to $4077/4096$ at $L = 12$. The maximum $|S|$ over the 50-point δ -grid rises monotonically from 2.199 at $L = 3$ to $|S| = 2.811$ at $L = 12$ (99.4% of the Tsirelson bound $2\sqrt{2}$, $C = 0.988$, at $\delta \approx 1.44\pi$); at $\delta = 0$, with the standard Fibonacci data, the $L = 12$ maximum is $|S| = 2.733$ (96.6%, $C = 0.932$). These $|S|$ values are signatures of nonseparability on the nonfactorized Fibonacci fusion space, not Bell violations in the Einstein–Podolsky–Rosen bipartite sense. The optimal length-12 sequence on the grid (at $\delta \approx 1.44\pi$) is $(\text{AM} \cdot \text{MB})^6$, where AM acts on Alice's qubit alone and MB on Bob's qubit conditioned on Alice's. Both values are exact within the model at their δ — 2.733 with the standard data at $\delta = 0$, 2.811 with the deformed data at $\delta \approx 1.44\pi$ — and were obtained by full enumeration over all 2^L sequences combined with the Horodecki bound $|S|_{\max} = 2\sqrt{1 + C^2}$ [7], as detailed in the companion paper Ref. [6]; see *Note on optimization* below for the v2.0 $\rightarrow$ v2.1 correction.

The CHSH-form values depend strongly on δ . When δ is varied (modifying the relative phase between fusion channels), the concurrence ranges from 0 to 0.99 over the $\delta \in [0, 2\pi)$ sweep ($N = 200$ points). Throughout this Fibonacci sweep, δ should be read as a representation-

deformation parameter rather than a physical anyon parameter: as verified numerically in the companion landscape analysis [[6], Appendix C], generic $\delta \neq 0$ breaks the Yang–Baxter/hexagon consistency of the braid representation, and only $\delta = 0$ realizes the standard Fibonacci modular tensor category. With *fixed* measure-

TABLE VI. Fibonacci anyon CHSH-form values vs. sequence length (values reproduced from Ref. [6] Tab. I via full enumeration over all 2^L sequences and the 50-point δ -grid; the $|S|$, C , % Tsir., δ^*/π , and sequence columns give the best sequence at a grid point δ^* that maximizes $|S|$ (the mirror-image point $6\pi/5 - \delta^*$, mod 2π , gives the same value), and $\delta = 0$ carries the standard Fibonacci data, where $L = 12$ gives $|S| = 2.733$; Horodecki bound $|S|_{\max} = 2\sqrt{1 + C^2}$ [7] from the maximal concurrence C). The last column counts how many of the 2^L sequences violate at $\delta = 0$; the preceding columns report the single best sequence. Sequence notation: $A = \text{AM}$ (gate on Alice's qubit), $B = \text{MB}$ (gate on Bob's qubit conditioned on Alice's).

L	$ S $	C	% Tsir.	δ^*/π	Sequence	Viol. ($\delta=0$)
3	2.199	0.457	77.7	0.08	ABB	4/8
4	2.225	0.488	78.7	1.40	ABAB	11/16
5	2.348	0.615	83.0	1.44	AABAB	26/32
6	2.433	0.693	86.0	1.48	AAABAB	57/64
7	2.536	0.780	89.7	0.12	ABBAABB	120/128
8	2.540	0.783	89.8	1.36	BABABAB	247/256
9	2.639	0.861	93.3	1.80	ABBBBABAB	502/512
10	2.696	0.904	95.3	0.08	ABBAABBAAB	1013/1024
11	2.784	0.968	98.4	1.44	ABABBBBABAB	2034/2048
12	2.811	0.988	99.4	1.44	ABABABABABAB	4077/4096

^a The original v2.0 table reported lower $|S|$ values for $L \leq 11$ from an X-Z-restricted angular optimizer; these have been replaced in v2.1 by the Horodecki-bound full enumeration of Ref. [6]. See *Note on optimization* below for the correction trail.

ment settings (from a numerical search at the grid maximum $\delta \approx 1.44\pi$), $|S|$ ranges from near zero (when settings are misaligned with the δ -dependent optimal basis) to 2.72, with $|S| > 2$ in 14% of the 200 δ values. When measurement settings are instead optimized at each δ via the full Bloch-sphere parametrization (the same methodology used in Table VI), the adapted-settings value is $|S| = 2.811$ at $\delta \approx 1.44\pi$ and again at $\delta \approx 1.76\pi$: the sweep takes the same value at both points, and the first of them is the one recorded as the Table VI optimum at $L = 12$. The two points are mirror images about $\delta = 8\pi/5$: complex conjugation maps the generators at δ to those at $6\pi/5 - \delta$ up to a global phase and leaves the concurrence unchanged, so the whole sweep is mirror-symmetric (to within 10^{-15} in the deposited data; $\delta = 6\pi/5$ reproduces the $\delta = 0$ value). On this finer 200-point sweep the maximum is $|S| = 2.816$, at $\delta \approx 1.43\pi$ and 1.77π , consistent with the grid value being a lower bound. With *adapted* settings via the Horodecki bound, $|S|$ never falls below 2.000, and $|S| > 2$ is achieved in 99.5% of the 200 δ values. Different values of δ thus require different detector calibrations for the maximum.

Unlike Ising anyons (Sec. IV D), Fibonacci anyons have no fermion parity restriction on local measurements. The full Pauli algebra is accessible in the fusion space, so measurements are not restricted by a parity rule as they are for Ising anyons, whose protocol saturates at $S = 2.000$; combined with the entangling gate MB, the d_{1b} circuit model gives CHSH-form values above 2, already at $\delta = 0$ ($|S| = 2.7334$). Among the models compared in Fig. 4, it is the only one that exceeds 2. Braiding alone with the two generators of this encoding does not exceed 2 (Ref. [6]), so this excess is a property of the circuit gates, not of braiding.

Note on optimization (added 2026-04-15, extended 2026-05-28). The length-12 grid maximum $|S| = 2.811$ ($\delta \approx 1.44\pi$) reported here supersedes an earlier estimate $|S| = 2.554$ obtained from the original angle-search optimizer (six random seeds plus a 9^4 grid). A subsequent full enumeration of winning sequences, combined with the analytic Horodecki bound $|S|_{\max} = 2\sqrt{1+C^2}$ [7], recovers $|S| = 2.811$ at $C = 0.988$ for the grid-optimal length-12 sequence $(AM \cdot MB)^6$ at $\delta \approx 1.44\pi$ (99.4% of the Tsirelson bound). The earlier value is therefore an optimization artifact, not a physical limit. The same enumeration shows that the two encodings differ in the group they generate: the two-generator d_{1b} circuit model generates a six-dimensional proper subgroup of $SU(4)$ (entangling but not universal), while the three-generator encoding is the standard B_4 braid representation on the five-dimensional fusion space (dense on each intermediate-charge sector [34]); under standard four-dimensional computational-subspace measurement both nevertheless reach comparable CHSH values near the Tsirelson bound, differing primarily in the minimum word length required ($L \approx 12$ vs. $L \approx 8$). Protocol-dependent differences — in particular the gap between native-5D CHSH and the 4D-projected value, and the

associated leakage statistics — are documented in the companion paper Ref. [6].

v2.1 update (2026-05-28). The same Horodecki-bound full enumeration has been applied to lengths $L = 3$ –11 in v2.1, replacing the original X-Z-plane restricted angular optimizer used in v2.0 (and earlier versions) for those rows of Tab. VI. The restricted optimizer reported systematically lower values (mean deficit -0.60 , maximum -1.04 at $L = 3$) and suboptimal winning sequences, because complex-phase R -matrices in the d_{1b} circuit model require measurement operators rotated outside the X-Z plane to saturate the Horodecki maximum. With the corrected values, the trend is monotonic in both $|S|$ and C from $L = 3$ to $L = 12$; the earlier “nonmonotonicity” interpretation in v2.0 was therefore an optimizer artifact and has been removed. See Ref. [6] Tab. I for the underlying enumeration.

Table VI update. Table VI, rows $L = 6$ and $L = 9$: the previously reported entries were the maxima of an earlier δ -sweep, not the maxima of the full Horodecki-bound enumeration. They are replaced by the enumeration maxima ($L = 6$: $|S| = 2.433$, $C = 0.693$, sequence AAABAB; $L = 9$: $|S| = 2.639$, $C = 0.861$, sequence ABBBBABAB). The remaining eight rows are unchanged. The underlying enumeration is deposited with this record.

IV. DISCUSSION

A. Summary of Established Results

We establish three main results at increasing levels of significance. First, topological sectors and CHSH values are not statistically independent in the $\mathbb{Z}_2$ lattice gauge theory at $\beta = 2.0$ ($I(T; S) = 0.0173$ bits on gauge-invariant observables after subtracting the null level, 95% interval $[0.0159, 0.0187]$, 21 of 21 seeds $>3\sigma$ against a circular-shift null whose measured false-positive level is 0.05 of 21). The decomposition of Table II shows that the association is carried by the excitation-number component of the sector label. The Wilson-loop component never dominates at any coupling examined, and its own contribution is bounded at 1.2×10^{-4} bits at $\beta = 1.0$, where the sector label is maximally mixed (1.9995 of 2 bits of entropy) and effectively fully sampled (3940 of 4000 measurements independent). The one coupling at which its interval excludes zero, $\beta = 2.5$, is the one with the poorest resolution among the couplings with a stated bound (73 of 4000 independent), and the apparent contribution grows as the resolution falls — the signature of a residual estimator bias rather than of physics. The origin of this mutual information is therefore not topological. Second, this mutual-information signal is a finite-size effect that decays toward zero with system size (Sec. III B). Third, and most significantly, the two-generator d_{1b} circuit model of Fibonacci anyons produces

CHSH-form values above 2: $|S| = 2.7334$ (96.64% of the Tsirelson bound) at $\delta = 0$, the point that realizes the standard Fibonacci modular tensor category, and $|S| = 2.8114$ (99.40%) at the maximum of the 50-point δ -grid, $\delta \approx 1.44\pi$, both at sequence length $L = 12$; as a grid maximum, 2.8114 is a lower bound on the maximum over δ . The values depend strongly on δ : with fixed detector settings, $|S|$ varies from near zero to 2.72 across $\delta \in [0, 2\pi)$.

The Fibonacci result contrasts with a hierarchy of negative results: $\mathbb{Z}_2$ classical models cannot violate Bell (Fine's theorem), abelian quantum anyons are locally invisible ($L \geq 3$ toric code), and Ising anyons saturate at $S = 2.000$ (fermion parity). Fibonacci anyons have no parity restriction, and the d_{1b} circuit model built from their F - and R -matrices, with an entangling gate, is the only model among those compared here that achieves CHSH-form values above 2.

The models tested in this work, with the CHSH value reported for each. Values are those of Tables VI, VII and VIII; the null results are listed because they carry the statement that the d_{1b} circuit model is the only one among those listed here to exceed 2.

Model	Setting	$ S $
XY 16×16 ($\beta=1.0$)	classical, 16×16 torus	0.55
XY + meas. coupling	classical, 16×16 torus	0.80
XY + $\rho(n a, b)$ weighting	classical, 16×16 torus	2.000
XY + transport phase	classical, 16×16 torus	0.44
$\mathbb{Z}_2$ gauge ($\beta=2.0$)	classical, 16×16 torus	0.13
$\mathbb{Z}_2$ gauge ($\beta=3.0$)	classical, 16×16 torus	1.34
Quantum TC $L=2$	exact diag., 8 qubits	1.41
Quantum TC $L=3$	exact diag., 18 qubits	0.00
Perturbed TC (h -scan)	exact diag., 8 qubits	0.00
Ising anyons (Majorana)	Majorana rep.	2.000
Fibonacci anyons ($L=12$)	circuit, 4-D, $\delta=0$	2.733
Fibonacci, deformed ($L=12$)	circuit, 4-D, $\delta \approx 1.44\pi$	2.811

B. Limitations

All results reported here are numerical simulations within idealized TQFT models, not laboratory experiments. The Fibonacci CHSH violation is computed in the ideal TQFT fusion space — a unitary, zero-temperature, zero-decoherence model. In a physical realization (e.g., $\nu = 12/5$ fractional quantum Hall state), thermal excitations, finite system size, and quasiparticle tunneling would reduce $|S|$ below the ideal value. In addition, the entangling gate of the circuit model is not a braid (Ref. [6]), so a physical realization would have to implement it by other means. The $\mathbb{Z}_2$ results are robust — the corrected value is unchanged under a tenfold longer run (+0.0173 against +0.0179 bits), while its significance grows from 10.5 to 74.8σ , as a real effect does and an estimator bias does not — but small: $I \sim 0.01$ – 0.04 bits. The XY model result is not robust across seeds. The elevated mutual information at $\beta = 2.5$ is carried by the excitation-number component of the label, not by a non-

trivial topological mechanism; the earlier reading of that regime as a near-frozen anyon sector rested on the superseeded engine.

C. Implications for Topological Interpretations

The Fibonacci result shows that, within the idealized TQFT model, the d_{1b} circuit model with the standard Fibonacci F and R data ($\delta = 0$) produces CHSH-form values above 2 with measurements on the two logical qubits ($|S| = 2.7334$). The sweep over δ is a deformation of the representation, not a set of topological sectors, so it does not by itself show a sector-dependent modulation; for Ising anyons, where the sector phase is physical, the next subsection gives the analytic form of that dependence.

D. Prediction: Sector-Dependent CHSH Correlations for Ising Anyons

For Ising anyons we present an analytical argument that non-Abelian topological order produces sector-dependent CHSH correlations. The Ising braiding R -matrix has two eigenvalues corresponding to the two fusion channels [12, 15]:

$$R_1 = e^{-i\pi/8}, \quad R_\psi = e^{3i\pi/8}. \quad (4)$$

On a torus, the topological sector modifies the relative phase between these channels. In sector δ , the R -matrix becomes $R_1 = e^{-i\pi/8}$, $R_\psi = e^{i(3\pi/8+\delta)}$, where δ depends on the anyon type threading the noncontractible loop [9, 15]. Here δ is a physical sector phase fixed by the threading anyon — a discrete, topologically determined label — rather than a freely tunable deformation parameter. Braiding two pairs of Ising anyons produces the entangled fusion state:

$$|\psi(\delta)\rangle = \cos \alpha |00\rangle + e^{i\delta} \sin \alpha |11\rangle, \quad (5)$$

where α is determined by the braiding sequence and $|0\rangle$, $|1\rangle$ label the fusion channels (vacuum and fermion respectively). For maximal braiding ($\alpha = \pi/4$), the concurrence is 1 for all δ — the entanglement is sector-independent. However, the CHSH value with *fixed* measurement settings is strongly sector-dependent:

$$S(\delta) = 2\sqrt{2} \cos^2(\delta/2) \quad (\text{settings optimized at } \delta=0). \quad (6)$$

We verify this numerically: with settings optimized for the vacuum sector ($\delta = 0$), S ranges from $2\sqrt{2} \approx 2.83$ (maximal CHSH violation) at $\delta = 0$ to $S = \sqrt{2} \approx 1.41$ at $\delta = \pi/2$, and $S = 0$ at $\delta = \pi$, following the $\cos^2(\delta/2)$ law. Eq. (6) is evaluated here as a continuous function of δ ; values between the discrete sector values do not correspond to a physical sector. Critically:

(i) **CHSH violation is preserved in all sectors** if measurement settings are adapted — $\max |S| = 2\sqrt{2}$ for

all δ . The topology does not create or destroy entanglement.

(ii) **With fixed settings, S depends on the sector** — producing $I(T; S) > 0$, the quantity measured for $\mathbb{Z}_2$ in Sec. III, here with CHSH values above 2.

(iii) **The sector rotates the optimal measurement basis** in the fusion space. Different sectors require different detector calibrations to see maximum violation. An observer who does not know the sector sees reduced S on average.

This constitutes a concrete analytical prediction, subject to the parity caveat below: in any experiment that creates and braids Ising anyons on a surface with non-trivial topology, the observed CHSH value must depend on the topological sector. The functional form — cos-modulation with period determined by the anyon type threading the handle — is parameter-free.

Caveat: Numerical verification via a Majorana fermion representation shows that the braid-then-fuse measurement protocol saturates at $S = 2.000$ exactly for Ising anyons, due to the fermion parity superselection rule: Pauli- X operators project to zero in the even-parity sector, restricting local measurements to the Z -basis. This limits Ising anyons to the classical bound when only topologically protected stabilizer operations—gates and measurements together—are available [27]. The prediction $S(\delta) = 2\sqrt{2}\cos^2(\delta/2)$ applies when measurements in the full fusion-space basis are available. Fibonacci anyons have no parity restriction, but the δ sweep of Sec. III G is a deformation, not a sector phase; the corresponding prediction for them is not derived here. A companion paper [26] constructs this noncommuting CHSH operator algebra locally in the Kitaev honeycomb model; as noted above, a genuine violation with topologically protected Ising operations requires nonstabilizer (magic) resources.

E. Related Work and Prior Art

The CHSH-violation construction presented here for Fibonacci anyons sits in a broader landscape of topological and nonstandard Bell tests. Earlier qualitative Bell-test constructions for non-Abelian anyons were proposed by Brennen, Iblisdir, Pachos, and Slingerland [16] for Fibonacci and Ising anyons; Chang, Chu, and Ma [17] studied Bell’s inequality in topologically ordered qubit models (Wen-plaquette). On Majorana platforms, Drummond, Kovalev, Hou, Shtengel, and Pryadko [18] gave an explicit CHSH protocol for Majorana wires, and Romito and Gefen [19] demonstrated nonlocal entanglement with six Majorana zero modes reaching the Tsirelson bound. Single-particle Bell-like inequality violations in neutron interferometry were reported by Hasegawa, Loidl, Badurek, Baron, and Rauch [20]; the contextuality-based program of Liu, Hu, Chen, Huang, Han, Li, Guo, and Cabello [21] provides a complementary view on the relation between nonlocality and contextuality. The present paper differs from these works in scope:

we report the mutual-information structure $I(T; S)$ between topological sectors and CHSH values across a hierarchy of lattice and anyonic models, including the finite-size scaling reported in Sec. III B, rather than a single Bell-test protocol on a fixed topological platform.

Independently, Xu, Ye, and You [22] have studied CHSH correlations on Fibonacci anyonic states using a three-copy fusion-sector observables construction (their Eq. (23)), reporting $|S| \approx 2.633$ on $|\tau, \tau; 1\rangle^{\otimes 3}$. Their setup differs from ours in both the ambient fusion space and the observable choice; a detailed setup-by-setup comparison appears in the companion paper [6].

Recent methodological work on braid-circuit verification by Amaral [23] and on minimal-depth Levin–Wen circuits by Bseiso *et al.* [24] provides additional methodological context for the F-symbol conventions and Yang–Baxter checks used here. A companion Leggett–Garg-inequality analysis using the same Fibonacci F - and R -matrices, with K_3 values reaching 99.998% of the Lüders bound $3/2$, is reported separately in Ref. [25].

F. Relation to Companion Work and Open Questions

Two of the directions previously listed here are addressed elsewhere. The complete enumeration of CHSH-form values over all words up to length $L = 12$ in the two-generator encoding is reported in the companion paper Ref. [6]; that paper makes no claim about the convergence of $|S|$ toward the Tsirelson bound for longer words. The temporal-correlation counterpart, using the same Fibonacci F - and R -matrices and Leggett–Garg-style protocols, is reported in Ref. [25]. The finite-size scaling of the $\mathbb{Z}_2$ sector–CHSH mutual information is settled within the present paper (Sec. III B), where it is shown to decay toward zero with system size. Two questions remain genuinely open. Whether a microscopic Hamiltonian realization — for example the Kitaev honeycomb model in its non-Abelian phase [15] — reproduces the sector dependence predicted for Ising anyons in Sec. IV D is not established here. And whether a physical sector phase produces a comparable dependence in other non-Abelian anyon models, including Fibonacci anyons and $SU(2)_k$ theories at level $k \geq 3$, is likewise not addressed by the present construction.

V. CONCLUSION

We have presented a systematic numerical investigation showing that the d_{1b} circuit model with the standard Fibonacci data ($\delta = 0$) produces CHSH-form values above 2 on the fusion space. The progression from classical $\mathbb{Z}_2$ gauge theory (a small, nontopological association: $I(T; S) = 0.0173$ bits on gauge-invariant observables, carried by the excitation number) through abelian quantum anyons (locally invisible at $L \geq 3$) to the d_{1b} cir-

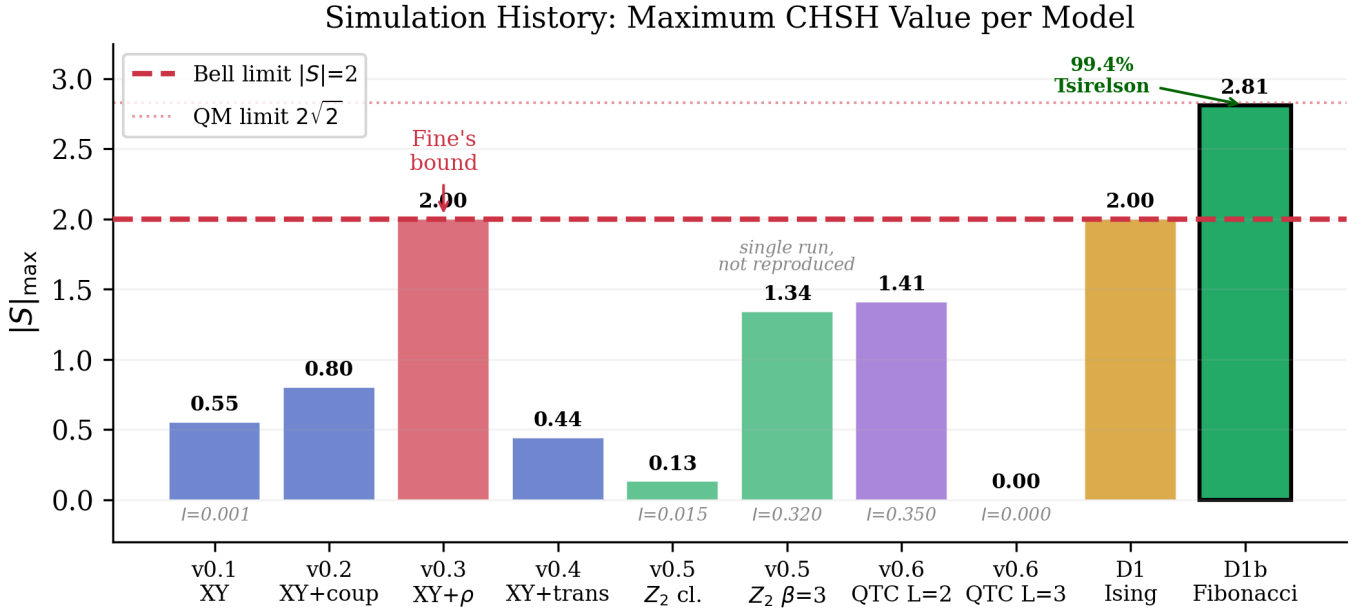


FIG. 4. Maximum CHSH value across all simulation models. The classical limit $|S| = 2$ (dashed red) is never exceeded in abelian models. The Fibonacci bar ($L = 12$) shows $|S| = 2.811$ (99.4% of the Tsirelson bound $2\sqrt{2}$; corrected value, see Note on optimization, Sec. III G), the maximum on the 50-point δ -grid of the deformed representation ($\delta \approx 1.44\pi$) and a lower bound on the maximum over δ ; at $\delta = 0$, the only point that realizes the standard Fibonacci modular tensor category, $|S| = 2.7334$.

cuit model built from the Fibonacci F - and R -matrices ($|S| = 2.7334$ at $\delta = 0$) shows that, among the models compared in Fig. 4, only this circuit model exceeds $|S| = 2$; braiding alone with the two generators of its encoding does not (Ref. [6]).

Over the Fibonacci deformation sweep the CHSH-form values depend strongly on δ (from near zero to 2.72 with fixed settings; above 2 at 99.5% of the δ values with adapted settings), and the concurrence varies from 0 to 0.99 (Sec. III G). For Ising anyons, where the sector phase is physical, the analytic argument of Sec. IV D predicts that the sector rotates the optimal measurement basis without changing the entanglement, subject to the parity caveat stated there. The $\mathbb{Z}_2$ mutual information is a finite-size, non-topological association, the Ising analysis identifies the fermion parity obstruction, and the Fibonacci result indicates that this obstruction is not universal. Together, they constitute a systematic investigation of CHSH correlations across abelian and non-Abelian models, with $I(\text{topology}; \text{CHSH})$ measured in the lattice and toric-code models.

NOTATION

Symbols used in this paper.

Symbol	Meaning
L	linear lattice size ($L \times L$ torus) or, for the anyon models, sequence length
β	inverse temperature of the lattice model

$\langle B_p \rangle$	single-plaquette expectation value
$I(T; S)$	mutual information between the topological sector label T and the CHSH value S
$ S $	magnitude of the CHSH functional
C	concurrence of the two-qubit state
θ	measurement angle, $M(\theta) = \cos \theta \sigma^z + \sin \theta \sigma^x$ (lattice models)
AM, MB	the two circuit gates: AM acts on Alice's qubit alone, MB on Bob's qubit conditioned on Alice's
δ (Ising)	physical sector phase fixed by the anyon threading the noncontractible loop, $R_\psi = e^{i(3\pi/8+\delta)}$
δ (Fibonacci)	representation-deformation parameter varied in the deformation sweep of Sec. III G; only $\delta = 0$ realizes the standard Fibonacci modular tensor category

DATA AND CODE AVAILABILITY

The seed statistics of Tables II and III (means, intervals and counts) are those of the corrected engine's per-seed values deposited in `data/null_calibration_results_vecprep_v2.json`. The deterministic analysis scripts and JSON output files deposited with this record reproduce the finite-size scaling of Tables IV and V (raw

run in `data/stufe2_results.json`, aggregated by `data/stufe2_analyse.py` into the per-cell data and model comparison of `data/stufe2_analyse.json`, the frozen-regime characterization, and Figs. 1(a) and 2; the systematic seed sets are {42, 10–29} for the decomposition and both rows of Table III, and 10–19 for the scaling cells. The per-seed entries of Table I are from the original analysis and have no deposited generating script; its $L = 16$ mean is independently reproduced by the deposited scaling run (Table IV, v05 column, to within 0.03σ). The original-analysis values in the footnote of Table III are reproduced, for the gauge-invariant route, by the deposited script `data/gauge_inv_plaquette_strip_recompute.py`; the independent re-implementation `data/gauge_inv_independent_route.py` reproduces the count of 21 of 21 seeds above 3σ . The gauge-fixed original values have no deposited generating script. The Fibonacci values of Table VI are reproduced by the deposited enumeration `data/fibonacci_enumeration.py` and its output `data/fibonacci_enumeration_results.json`, which cover all ten lengths $L = 3$ –12 and import the gate definitions read-only from the companion record Ref. [6] (concept DOI [10.5281/zenodo.19601352](https://doi.org/10.5281/zenodo.19601352)); the δ sweep of one selected sequence per length for $L = 3, 6, 8, 10, 12$ is held in that record’s `s_delta_sweeps.json`, which is deposited there and is not part of the present package. The δ sweep behind the fixed- and adapted-settings figures of Sec. III G — 14% and 99.5% of the 200 sampled values, and the range up to 2.72 — is held in `data/sector_range_fixed_settings.json`, first deposited with version 2.0 of this record ([10.5281/zenodo.19656928](https://doi.org/10.5281/zenodo.19656928)) and included here. Its Horodecki column is reproducible from the deposited concurrence column through $|S|_{\max} = 2\sqrt{1 + C^2}$; the fixed-settings angles come from a numerical search whose generating script is not deposited, so those entries record what the deposited run yielded rather than what the method yields on a re-run. The Ising saturation $S = 2.000$ follows analytically from the fermion parity superselection rule and is not a numerical result. The remaining entries of Table VII and the corresponding bars of Fig. 4 record earlier model iterations and are kept as history. All files are archived on Zenodo under the concept DOI [10.5281/zenodo.19600752](https://doi.org/10.5281/zenodo.19600752) (which always resolves to the latest version). Paper, figures, and data are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; the deposited source code is released under the Apache License 2.0 (see LICENSE and LICENSE-CODE).

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances

that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI’s.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

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Appendix A: Simulation Parameters and Negative Results

All classical simulations use $L = 16$ with Metropolis MC (single-site for XY, single-edge for $\mathbb{Z}_2$). The legacy engine and the validation scripts thermalize for 2000 sweeps and take 4000 measurements per run (the frozen-regime characterization: 40 000 measurements); an exploratory β scan of the legacy engine uses 1000/2000. The run lengths of the corrected engine, which grow with L in the scaling cells, are listed in the deposited record. MI permutation test: 500–2000 shuffles. Detectors at (4, 4) and (12, 12). Quantum toric code: exact diagonalization via projector method.

TABLE VII. Complete simulation history. The $\mathbb{Z}_2$ $\beta=3.0$ row is a single early frozen-regime run, kept as history; it is not reproduced by the deposited systematic seed set (Table II). The $\mathbb{Z}_2$ rows at $\beta=2.0$ give the original analysis on a superseded engine; corrected values in Sec. III.

Model	$ S _{\max}$	I (bits)	σ	Key finding
XY 16×16 ($\beta=1.0$)	0.55	0.001*	1.5*	Signal not robust (4/10 seeds)
XY + meas. coupling	0.80	–	–	Coupling adds noise
XY + $\rho(n a, b)$ weighting	2.000	–	–	Fine bound reached exactly
XY + transport phase	0.44	–	–	$\cos \theta$ form, 17% amplitude
$\mathbb{Z}_2$ gauge ($\beta=2.0$)	0.13	0.015	10.8	Original: 9/10 seeds $>2\sigma$; corrected engine: 10/21
$\mathbb{Z}_2$ gauge-inv. ($\beta=2.0$)	–	0.018	14.2	Confirmed: 21/21 $>3\sigma$
$\mathbb{Z}_2$ gauge ($\beta=3.0$)	1.34	0.320	590	Frozen regime (trivial)
Quantum TC $L=2$	1.41	0.350	1595	Finite-size artifact
Quantum TC $L=3$	0.00	0.000	0	Abelian: locally invisible
Perturbed TC (h -scan)	0.00	–	–	Perturbation destroys correlations
Ising anyons (Majorana)	2.000	–	–	Fermion parity: $S \leq 2$ locally
Fibonacci anyons (len=12)	2.811	–	–	CHSH-form values above 2: 99.4% Tsirelson[†] at $\delta \approx 1.44\pi$; $S = 2.733$ at $\delta=0$

^a * Not robust across independent seeds. – Not reported for this model. [†] Updated from full enumeration; see Note on optimization, Sec. III G. Bold: primary results.

TABLE VIII. Negative results.

Test	Result	Implication
Wilson-loop CHSH ($L=3$)	All correlations zero	Stabilizers + abelian = trivial
Fourier mode $\propto n$	Dominant $k=1$ for all $ n $	No frequency shift with sector
Resonance at $\theta=\pi/n$	Ratio 1.2–1.4 (weak)	Suggestive but not significant
XY seed variation (10 runs)	4/10 $>2\sigma$	XY not robust for claim
Perturbed TC (all h)	$ S =0$ for $h>0$	Perturbation destroys $L=2$ corr.
Ising braid-then-fuse	$S=2.000$ exactly	Fermion parity superselection

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